Experiment :- Study of LAN Topology, Network devices and Address Configuration

Objective :-

1. To study the LAN Topology used in College Campus.
2. To identify networking devices & understand their function.
3. To analyse wired & wireless Communication method used in the network.
4. To measure speed of network for wired & wireless Connections.
5. To obtain MAC, IP Address & subnet mask.

→ Problem statement - 1

. Draw & Identify LAN Topology of College Campus & clearly label all components such as roofer, switches & etc.

Soln → The College Campus network observed this study consist of :

* Layer-3 Core switches in the Corner room.
* Managed by PDE & switches.
* wireless Access Point (APs)

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* Ethernet Port in classroom & Laboratories.

→ Network Topology

Hybrid Hierarchical Topology which Combines
(1) Tree Topology
(2) Extended Star Topology
(3) Ring Topology (server room)

→ The Network Hierarchical order:-

- Core layer
- Distribution layer
- Access layer

→ Core layer -

Core layer Contains server room & server room have Firewall, Router, 2-L3+1 switches, Patch Rack, server, inverter etc.

→ Distribution layer -

* Distribution layer Contains 4-switches.
* The 4 switches are -
  (1) Main build → R7 (Room no-213, 2nd floor)
  (2) Main build → R1 (Room no-301, 3rd floor)
  (3) Main build → R6 (Room no-113, 1st floor)
  (4) Main build → R2 (Room no-201, 2nd floor)
* This 4-switches Contains —
(1) 18 Port Gigabit Easy Smart switch with 16-Port.

TL-SG1218MPE

(2) SG3428XMP → 24 Port Gigabit & 4-Port 10GE SFP+, L2+ Managed Switches with 24 Port POE+.

→ Now Let's discuss these 2 above Components →

(1) TP-Link TL-SG1218MPE

* Likely used for: CCTV Cameras, wifi access Point etc.
* 18 Port Gigabit "Easy Smart" switch.
* 16 Port support POE+

(2) TP-Link Omada SG3428XMP

* Higher-end managed Layer 2+switch.
* 24POE+ Port, Uplinks, 10G SFP
* usually acts as a main distribution for each floor.

→ Access Layer

* Access layer Contain 3-switches i.e
  (1) Main build R1-[Room no.-301, 3rd floor].
  (2) Main build R4-[Room no.-05, Ground floor].
  (3) Main Build R5-[Room no.-18, Ground floor].

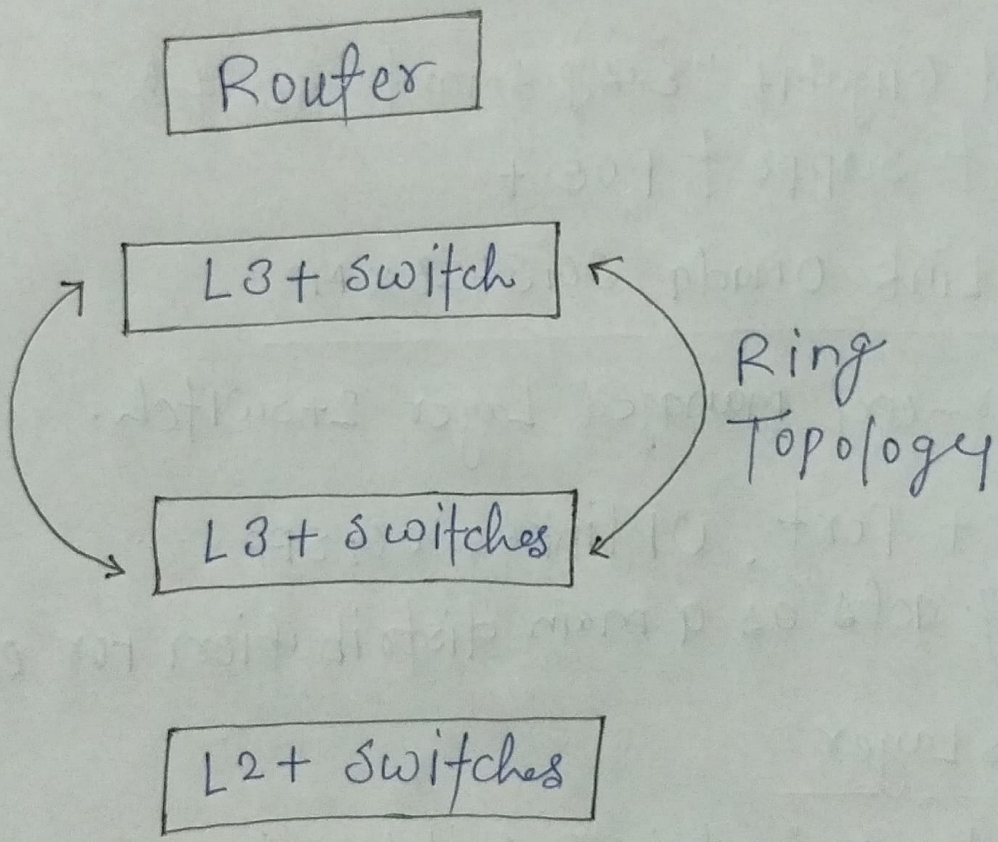
Every switches act as access layer which helps to provide network connectivity to all Classroom, Lab, Facuelty Room, etc.

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* Access layer directly connect :

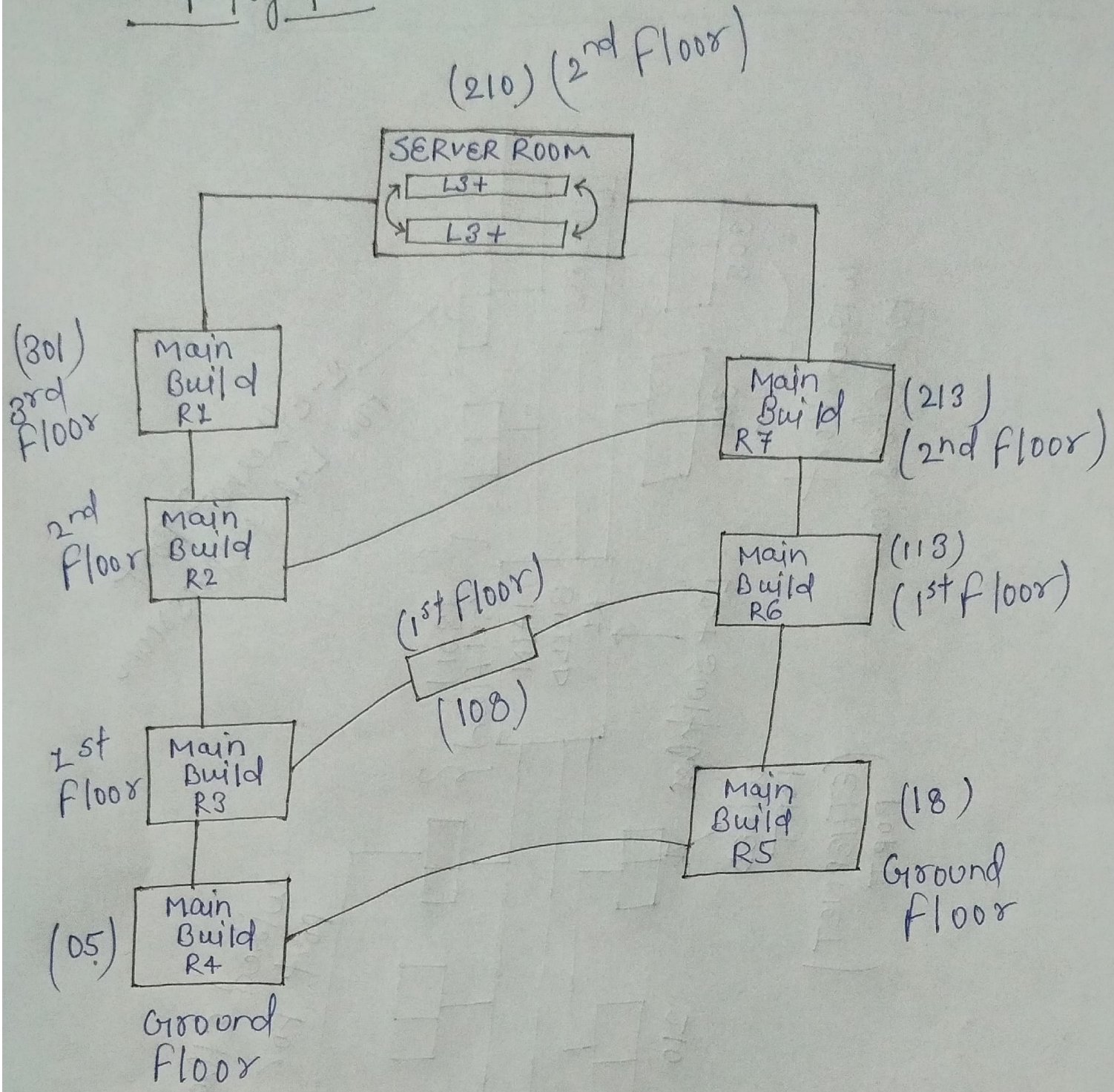
1. Ethernet Port
2. wireless AP
3. Lab system
4. outdoor wireless AP (ex-018)

Server Room Architecture / Topology



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Topology



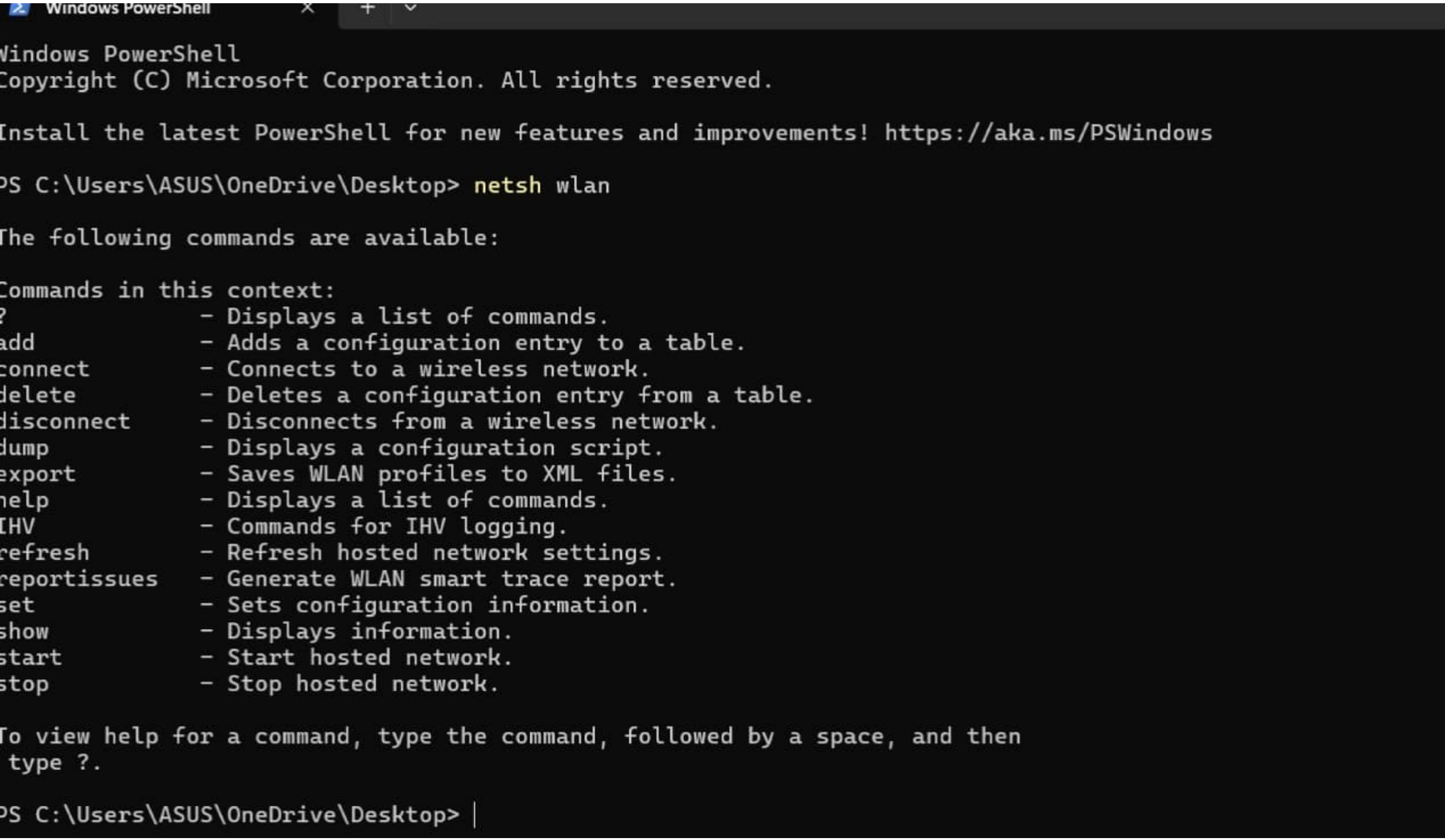
→ R1, R7, R2, R6 → Distribution layer

→ server room → Core layer

→ R1, R2, R3, R4, R5, R6, R7 → Access layer

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Access Layer Switch (3rd Floor-301)



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```
S C:\Users\ASUS\OneDrive\Desktop> ipconfig

indows IP Configuration


thernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::be00:7633:be99:1393%38
   IPv4 Address. . . . . . . . . . . : 10.95.116.193
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 10.95.116.213

ireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

ireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

ireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::dca1:e0c3:dcb2:1247%9
   IPv4 Address. . . . . . . . . . . : 10.129.238.188
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 10.129.238.195

thernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
S C:\Users\ASUS\OneDrive\Desktop>
```

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```
PS C:\Users\ASUS\OneDrive\Desktop> getmac

Physical Address    Transport Name
=================== ==========================================================
70-CD-0D-34-9A-A4   \Device\Tcpip_{56FB7643-FFDC-4ACE-9594-641147334FDF}
70-CD-0D-34-9A-A8   Media disconnected
PS C:\Users\ASUS\OneDrive\Desktop>
```

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→ Difference bet^n wired & wireless speed

Sol^n :-

wired	wireless
* faster Communication	* Slower Communication
* Stable Connection	* Affected by distance/ Interference
* Low latency	* Higher latency
* Dedicated Physical Medium	* shared radio Medium

→ Ethernet Provides direct physical Communication using network cables, resulting in higher speed & stable Performance.

→ Wireless Connections uses radio waves which are affected by interference, distance, obstacles & multiple Connected user.

(Q.N) Role of subnet mask in network?

→ Sol^n :- A subnet mask identifies the network Portion & host Portion of an IP address. It helps devices determine whether another device belongs to the same network or different network.

→ It is important for:

(1) Network segment
(2) Efficient routing
(3) Commonication b/w devices.

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→ Observation

During the study of the college network infrastructure:-

(A) A Centralized server room was used as network backbone.

(B) Two L3 switches were connected in redundant Cyclic structure.

(C) Managed TP-Link POE+ switches were used in network cabinets.

→ Advantage of existing Network

* scalable architecture
* Centralized management
* Supports POE + devices
* Easy expansion of floor & classroom.
* Support both wireless & wired.
* Redundant link for failover.

→ Conclusion

The observed campus LAN follow & hybrid hierarchical topology Consist of Tree, Extended Star, Ring. The network uses managed POE switches, wireless access Point, Ethernet Port & Centralized server infrastructure to Provide efficient wired & wireless Communication across the Campus.

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